Conditioning Protein Sequence Generation via Pattern Multiplicity in the Modern Hopfield Energy

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Abstract

Training-free protein sequence generation via stochastic attention produces plausible family members from small alignments, but treats all stored patterns equally and cannot direct generation toward a user-specified functional subset—for example, sequences known from experimental screens to bind a particular target. We show that assigning multiplicity weights r_k to stored patterns defines a tilted Hopfield energy whose Langevin dynamics sample from a known Boltzmann distribution—requiring only a log-multiplicity bias on the softmax logits and no retraining. A single scalar, the multiplicity ratio $\rho = r_{\text{designated}}/r_{\text{background}}$, continuously interpolates between unconditioned generation ($\rho = 1$) and hard subset curation ($\rho \to \infty$). We characterize the phase transition $\beta^*(\rho)$ as a function of the effective pattern count $K_{\text{eff}}(\mathbf{r})$ and identify a family-dependent calibration gap between energy-level conditioning and decoded sequence phenotype. The gap is predicted quantitatively by the Fisher separation of functional subsets in PCA space: families with separation index $S > 0.3$ achieve near-perfect phenotype transfer via multiplicity weighting alone, while families with $S < 0.15$ require hard curation. Experiments on Kunitz, SH3, and WW domains validate the theory across a tenfold range of separation indices ($S = 0.11$–0.34) and calibration gaps ($\Delta = 0.01$–0.63), with $R^2 \approx 1.0$ on the predicted linear relationship.

Significance Statement. Protein engineers routinely isolate a handful of functional binders—via phage display, yeast surface display, or functional selections—but lack a systematic way to expand these small sets into diverse candidate libraries without retraining a generative model. We show that a single scalar parameter added to the softmax logits of a Hopfield-energy sampler continuously conditions generation from "any family member" to "only binders," with no training, no GPU, and no architectural changes. The effectiveness of this conditioning is predicted *a priori* by a cheap geometric property of the alignment, giving practitioners a decision rule before any sequences are generated.

1 Introduction

Stochastic attention (SA) generates protein sequences from small family alignments by treating the modern Hopfield energy as an exact score function for Langevin dynamics [1]. Given a memory matrix of K PCA-encoded family members, SA samples from the corresponding Boltzmann distribution without training, producing sequences that preserve family-level amino acid composition, per-position conservation, and predicted three-dimensional fold. The method requires only two matrix-vector products per iteration ($\mathcal{O}(dK)$ cost), runs on a laptop in seconds, and is designed for the scarce-data regime—families with tens to hundreds of members—where learned generative models lack sufficient

Preprint.

training signal. A key limitation of the current framework, however, is that it treats all stored patterns equally. Every family member contributes the same weight to the energy landscape, so the Langevin chain explores the full family distribution without preference for any functional subset. In practice, experimentalists often want something more specific: not just "a plausible Kunitz domain," but a Kunitz domain that *inhibits trypsin*; not just "a plausible SH3 domain," but one that binds a particular proline-rich motif. Binding is relational—it depends on both the generated protein and the target—and family alignments encode fold constraints but not target specificity. The question is: *how can we condition the generative process on a functional subset without retraining?*

A seemingly simple answer is to *curate* the memory matrix, restricting it to sequences that a practitioner has identified—through experimental screens, literature annotation, or functional assays—as belonging to the functional subset of interest. SA then generates new sequences in their image. We show that this strategy works remarkably well: even three designated input sequences suffice for perfect phenotype transfer in the Kunitz domain. But hard curation discards all remaining family members, losing the fold-level constraints they provide. A more principled approach would retain the full family in the memory while continuously upweighting the designated subset. The key observation we exploit is that pattern multiplicity in the Hopfield energy provides exactly this capability. Assigning a multiplicity weight $r_k > 0$ to each stored pattern $\mathbf{m}_k$ defines a tilted energy landscape whose score function remains analytic—the only modification to the SA update is adding $\log r_k$ to each logit before the softmax. A single parameter, the multiplicity ratio $\rho = r_{\text{designated}}/r_{\text{background}}$, parameterizes a continuous family of Boltzmann distributions that interpolate between unconditioned generation ($\rho = 1$, standard SA) and hard subset curation ($\rho \to \infty$). The memory matrix stays compact at $d \times K$; no columns are duplicated, no parameters are trained, no architectural changes are needed. The method is agnostic to what the designated subset represents—binding, thermostability, organism of origin—it simply biases the sampler to spend more time near the patterns the user points to.

We develop the theory of multiplicity-weighted SA and test it on three protein families—Kunitz domains (PF00014), SH3 domains (PF00018), and WW domains (PF00397)—spanning a range of family sizes ($K = 55$–420), PCA dimensions ($d = 46$–186), and functional-subset geometries. Our main contributions are: (i) the multiplicity-weighted Hopfield energy and its Langevin dynamics, with a characterization of the phase transition $\beta^*(\rho)$ as a function of the effective pattern count $K_{\text{eff}}(\mathbf{r})$; (ii) a decomposition of the calibration gap between energy-level conditioning and decoded phenotype into three independently measurable layers (attention, PCA reconstruction, argmax decoding), revealing that the gap arises entirely from the PCA encoding; and (iii) a cross-family validation showing that the Fisher separation index S of functional subsets in PCA space quantitatively predicts the calibration gap magnitude ($R^2 \approx 1.0$ across three families spanning a tenfold range of S). These results provide a practitioner's decision rule: compute S, choose between multiplicity weighting and hard curation, and generate.

2 Results

We first asked whether restricting the memory matrix to a user-designated functional subset transfers the subset's phenotype to generated sequences. As a test case, we used Kunitz domains (PF00014; $K = 99$ sequences, $L = 53$ aligned positions), a family of serine protease inhibitors whose binding specificity is known from decades of biochemistry to depend on the P1 residue at the primary contact site with the target protease: Lys or Arg at P1 confers strong trypsin inhibition, while other residues (Gly, Gln, Asn) do not [**?**]. Using this external knowledge, we designated the 32 sequences with K/R at P1 as the functional subset ($K_{\text{des}} = 32$) and the remaining 67 as background ($K_{\text{bg}} = 67$), built separate memory matrices from each group, found β^* for each via entropy inflection, and generated 930 sequences per condition (30 chains $\times$ 5,000 steps, thinned after burn-in). The result was unambiguous: 100% of designated-subset-conditioned sequences had K/R at P1; 0% of background-conditioned sequences did. Generation from the full family produced 39% K/R, matching the natural proportion ($32/99 = 32\%$) within sampling noise. All generated sequences had 100% valid residues, and KL divergence of amino acid composition remained below 0.012 for all conditions. The designated phenotype was inherited perfectly; the fold-level composition was preserved (Fig. **??**).

We next assessed the minimum number of designated input sequences required for reliable phenotype transfer by subsampling $K_{\text{des}} \in \{3, 5, 8, 10, 15, 20, 25, 32\}$ from the Kunitz K/R subset (3 replicates each), building curated memory matrices, and generating sequences. P1 phenotype fidelity was

100% at *every* sample size, including $K_{\mathrm{des}} = 3$. The cost of small input sets was reduced diversity: pairwise sequence diversity increased from 0.36 ($K_{\mathrm{des}} = 3$) to 0.50 ($K_{\mathrm{des}} = 32$), and KL divergence decreased from 0.047 to 0.009 over the same range. These trends are expected: fewer stored patterns constrain the PCA subspace, reducing the volume accessible to the Langevin chain. Phenotype fidelity, however, was never compromised (Fig. **??**). This result validates the "small-set amplifier" framing: a practitioner with as few as 5–10 experimentally characterized sequences—for instance, hits from a phage display screen—can generate hundreds of novel candidates that maintain the input phenotype while introducing substantial sequence diversity (pairwise diversity ≥ 0.45 for $K_{\mathrm{des}} \geq 10$).

We then turned to multiplicity-weighted generation, sweeping ρ from 1 (standard SA) to 1,000 on the full Kunitz family ($K = 99$). For each ρ, we computed the multiplicity vector $\mathbf{r}$, found $\beta^*(\mathbf{r})$ via entropy inflection on the weighted attention distribution, and generated 930 sequences (30 chains). The attention weight on designated patterns tracked f_{eff} with negligible deviation ($< 0.3\%$) at all ρ values, confirming Proposition 1: the multiplicity weights tilt the energy landscape exactly as intended. The observed P1 K/R fraction increased monotonically from 0.39 ($\rho = 1$) to 0.63 ($\rho = 500$), substantially exceeding the natural proportion but falling short of the $f_{\mathrm{eff}} = 0.996$ predicted by the energy-level conditioning. Sequence diversity decreased modestly from 0.56 to 0.51 over this range; KL divergence increased from 0.007 to 0.017 (Fig. **??**).

The phase transition β^* shifted from 4.3 at $\rho = 1$ ($K_{\mathrm{eff}} = 99$) to 10.1 at $\rho = 1{,}000$ ($K_{\mathrm{eff}} = 32.1$), consistent with fewer effective patterns requiring higher inverse temperature to produce an ordered phase. The entropy curves at different ρ values formed a family of sigmoidal profiles with systematically rightward-shifted inflection points, confirming that $\beta^*(\mathbf{r})$ depends on $K_{\mathrm{eff}}(\mathbf{r})$ rather than the raw pattern count K (Fig. **??**).

To understand why f_{obs} lagged f_{eff}, we measured all three layers of the signal chain (Eq. 6) independently across a fine ρ sweep ($\rho \in [1, 1000]$, 17 values, 5 replicates each). At each sampling step, we recorded: (i) the total softmax attention weight on designated patterns ($\bar{a}_{\mathrm{des}}$), (ii) the continuous K/R probability at P1 in the PCA-reconstructed one-hot vector (f_{soft}), and (iii) the hard-decoded residue identity (f_{obs}). The decomposition revealed a striking structure. The attention gap was $\Delta_{\mathrm{attn}} \approx 0$ across all ρ values ($< 0.3\%$), confirming that the energy-level conditioning is exact. The PCA gap was the dominant component: f_{soft} was effectively flat at ~ 0.27 regardless of ρ, varying by less than 0.04 across a thousandfold range of multiplicity ratios. The argmax gap was negative (≈ -0.25), indicating that discrete decoding partially recovered signal lost in the PCA reconstruction (Fig. **??**). The flatness of f_{soft} implies that the PCA encoding maps designated and background sequences to overlapping regions at the P1 position: even when 99.8% of attention weight is on the designated patterns, the continuous one-hot reconstruction cannot resolve K vs. G at P1 because this variation is not well-captured by the top $d = 80$ principal components. The Fisher separation index for the Kunitz designated/background split was $S = 0.20$, confirming moderate but incomplete geometric separation.

We repeated the multiplicity analysis on two additional families—SH3 domains (PF00018; $K = 55$, split by conserved Trp at the binding groove) and WW domains (PF00397; $K = 420$, split by specificity-loop residue)—yielding three families with distinct PCA-space geometries (Table **??**). The calibration gap decreased monotonically with increasing separation index: WW domains ($S = 0.11$, functional subsets heavily interleaved in PCA space) exhibited the largest gap ($\Delta = 0.63$); Kunitz domains ($S = 0.20$) an intermediate gap ($\Delta = 0.37$); and SH3 domains ($S = 0.34$, well-separated functional subsets) a near-zero gap ($\Delta = 0.01$, $f_{\mathrm{obs}} = 0.989$ at $f_{\mathrm{eff}} = 0.999$). A linear fit yielded $\Delta \approx 0.84 - 2.5\,S$ with $R^2 \approx 1.0$. The attention weights tracked f_{eff} perfectly in all three families, confirming that the multiplicity-weighted energy is exact regardless of the downstream calibration gap. Hard curation achieved 100% phenotype transfer in all three families, regardless of separation index—as expected, since curated memory matrices are built from designated-subset-only PCA, which by construction captures subset-specific variation in its leading components (Fig. **??**).

Finally, we asked whether increasing β beyond β^* could push more phenotype signal through the PCA bottleneck. At fixed ρ, sweeping β/β^* from 0.5 to 3.0 revealed a consistent tradeoff across all three ρ values tested ($\rho = 10, 50, 200$): higher β increased P1 K/R fraction at the cost of reduced diversity. At $\rho = 200$ and $\beta = 3\beta^*$, P1 K/R reached 0.71 (versus 0.61 at β^*), while diversity dropped from 0.53 to 0.46. This provides a second continuous dial: the practitioner can tune $(\rho,\ \beta/\beta^*)$ jointly to navigate the fidelity–diversity tradeoff.

3 Discussion

We have shown that pattern multiplicity in the modern Hopfield energy provides a principled, training-free mechanism for conditioning protein sequence generation on functional subsets. The theoretical framework is exact: multiplicity weights tilt the Boltzmann distribution through a simple logit bias, and the resulting Langevin dynamics sample from a known distribution with no approximation. A single scalar ρ continuously interpolates between unconditioned generation and hard subset curation, and the practitioner can compute the required ρ for any target effective designated fraction directly from the family composition (Eq. 4). On the Kunitz domain, curated memory achieved 100% phenotype transfer with as few as three designated input sequences; multiplicity-weighted generation on the full family increased phenotype fidelity from the natural proportion (32%) to 63% at $\rho = 500$; and the cross-family validation on Kunitz, SH3, and WW domains revealed a quantitative relationship between PCA-space geometry and conditioning effectiveness.

The most informative finding is the calibration gap and its decomposition. The gap is not a failure of the multiplicity-weighted energy—the attention weights track f_{eff} with $< 0.3\%$ deviation across all families and ρ values tested—but a well-characterized property of the PCA encoding that compresses $20L$-dimensional sequence space into d principal components. When the designated and background subsets overlap in PCA space (low S), the energy-level tilt does not propagate to the decoded residue identity; when they separate (high S), it does. The cross-family validation produced a linear relationship between S and Δ with $R^2 \approx 1.0$, suggesting that the separation index is a reliable *a priori* predictor of conditioning effectiveness. This yields a practitioner's decision rule: compute S (a cheap operation requiring only PCA and pairwise cosine similarities, no generation); if $S > 0.3$, multiplicity weighting preserves fold-level constraints from the full family while achieving near-perfect phenotype transfer (SH3 domains); if $S < 0.15$, hard curation is necessary because the PCA bottleneck attenuates the signal too severely (WW domains); the intermediate regime ($0.15 < S < 0.3$, Kunitz domains) admits a tradeoff controlled by $(\rho,\ \beta/\beta^*)$.

The fact that hard curation achieves perfect phenotype transfer with as few as three designated input sequences is practically significant. Many experimental campaigns—phage display, yeast surface display, functional selections, or even literature curation—produce exactly this kind of small, functionally characterized set. The practitioner brings external knowledge ("these sequences bind my target," "these variants are thermostable," "these hits came from my screen") and SA converts this small set into a large, diverse candidate library while preserving the input phenotype. This "small-set amplifier" capability complements high-throughput screening: the screen identifies a few hits, SA expands them computationally, and the expanded library can be fed back into a second round of selection or directly into functional assays. The method itself is agnostic to the meaning of the designation—it biases the sampler toward the designated patterns regardless of whether the underlying phenotype is binding, stability, expression, or any other property. The scaling study confirms that even $K_{\text{des}} = 3$ suffices for phenotype transfer; diversity improves with more input sequences but saturates near $K_{\text{des}} = 20$.

Several limitations should be acknowledged. The cross-family regression rests on three points; additional families are needed to confirm the linearity of the S–Δ relationship and to establish its generality beyond single-position phenotypic markers. Our functional splits are defined by single marker positions (P1 in Kunitz, Trp in SH3, a specificity-loop residue in WW); multi-position phenotypes—where binding specificity depends on coordinated residue identities at several positions—may exhibit different calibration behavior. We have not experimentally validated the binding activity of generated sequences; the phenotype fidelity reported here is a necessary but not sufficient condition for functional binding. Computational binding affinity predictions (e.g., via AlphaFold-Multimer or FoldX) could provide an intermediate validation tier between sequence-level markers and experimental assays. Finally, we tested interface-weighted PCA encoding as a strategy to close the calibration gap, but found it neutral at low weights and destructive at high weights; the PCA bottleneck reflects the geometric entanglement of functional and structural variation in the family, not the absence of interface-position components.

Looking forward, the multiplicity framework opens several directions. The binary split (designated/background) generalizes naturally to multi-class conditioning by assigning different multiplicities to different functional categories. The PCA bottleneck that limits conditioning effectiveness might be reduced by alternative sequence encodings—learned representations from protein language models, for instance, may better separate functional subsets than linear PCA. The β lever suggests that

adaptive temperature schedules, increasing β during sampling to sharpen the multiplicity bias, could improve phenotype transfer without permanently sacrificing diversity. These extensions remain within the analytic, training-free framework that makes stochastic attention practical for the scarce-data regime.

4 Materials and Methods

4.1 From Hopfield Energy to Multiplicity-Weighted Score Function

Let $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$ be a memory matrix whose columns are unit-norm, PCA-encoded protein sequences from a family alignment of K members, each represented in d dimensions. The modern Hopfield energy [2] at state $\boldsymbol{\xi} \in \mathbb{R}^d$ is $E(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{\xi}\|^2 - \frac{1}{\beta}\log\sum_k \exp(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi})$, where $\beta > 0$ is an inverse temperature. The score function $\nabla_{\boldsymbol{\xi}} \log p_\beta(\boldsymbol{\xi}) = \mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi}) - \boldsymbol{\xi}$ is *exact* and computed by a single attention operation; applying the unadjusted Langevin algorithm (ULA) with step size $\alpha \in (0, 1)$ yields the stochastic attention update [1]:

$$\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\,\mathrm{softmax}\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t\big) + \sqrt{2\alpha/\beta}\,\boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d). \tag{1}$$

We extend this framework to condition generation on a functional subset by assigning a multiplicity weight $r_k > 0$ to each stored pattern. The *multiplicity-weighted* Hopfield energy is

$$\boxed{E_{\mathbf{r}}(\boldsymbol{\xi}) = \tfrac{1}{2}\|\boldsymbol{\xi}\|^2 - \tfrac{1}{\beta}\log\sum_{k=1}^{K} r_k\,\exp\big(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi}\big),} \tag{2}$$

where $\mathbf{r} = (r_1, \ldots, r_K)^\top$. When $r_k = 1$ for all k, Eq. (2) reduces to the standard energy. Patterns with higher multiplicity deepen the energy wells in their vicinity; this is equivalent to storing r_k copies of pattern $\mathbf{m}_k$ without the $\mathcal{O}(d \cdot \sum_k r_k)$ memory cost.

Proposition 1 (Score function)**.** *The score function of* $p_{\mathbf{r}}(\boldsymbol{\xi}) \propto \exp(-\beta\,E_{\mathbf{r}}(\boldsymbol{\xi}))$ *is* $\nabla_{\boldsymbol{\xi}} \log p_{\mathbf{r}}(\boldsymbol{\xi}) = \mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi} + \log\mathbf{r}) - \boldsymbol{\xi}$*, where* $\log\mathbf{r} = (\log r_1, \ldots, \log r_K)^\top$ *is the elementwise log-multiplicity.*

The proof follows from the identity $r_k\,\exp(a_k) = \exp(a_k + \log r_k)$: the weights are absorbed into the softmax logits. The resulting Langevin update is

$$\boxed{\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\,\mathrm{softmax}\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t + \log\mathbf{r}\big) + \sqrt{2\alpha/\beta}\,\boldsymbol{\epsilon}_t.} \tag{3}$$

The *only* change from Eq. (1) is the addition of $\log r_k$ to each logit before the softmax. The memory matrix $\mathbf{X}$ is unchanged, the noise schedule is unchanged, and the sampler retains its $\mathcal{O}(dK)$ per-step cost.

4.2 Multiplicity Ratio and Phase Transition

For a binary functional split (designated subset vs. background), we parameterize the multiplicity vector by a single scalar $\rho = r_{\text{designated}}/r_{\text{background}}$, setting $r_k = \rho$ for patterns in the user-designated subset and $r_k = 1$ otherwise. The effective designated fraction—the share of the softmax probability mass attributable to designated patterns when all patterns have equal similarity to $\boldsymbol{\xi}$—is $f_{\text{eff}}(\rho) = K_{\text{des}}\rho/(K_{\text{des}}\rho + K_{\text{bg}})$, where K_{des} and $K_{\text{bg}} = K - K_{\text{des}}$ are the number of designated and background patterns. At $\rho = 1$, $f_{\text{eff}} = K_{\text{des}}/K$; as $\rho \to \infty$, $f_{\text{eff}} \to 1$. Inverting gives the multiplicity ratio needed for a target fraction f:

$$\rho(f) = \frac{f \cdot K_{\text{bg}}}{K_{\text{des}}\,(1-f)}. \tag{4}$$

The standard SA framework exhibits a phase transition at critical inverse temperature β^* that depends on the PCA dimension d and the number of stored patterns. Under multiplicity weighting, the relevant quantity is the effective number of patterns $K_{\text{eff}}(\mathbf{r}) = (\sum_k r_k)^2/\sum_k r_k^2$, which equals K when all r_k are equal and converges to K_{des} as $\rho \to \infty$. Empirically, β^* increases monotonically from 4.3 ($\rho = 1$, $K_{\text{eff}} = 99$) to 10.1 ($\rho = 1000$, $K_{\text{eff}} = 32.1$) in the Kunitz domain, consistent with the system becoming harder to order as fewer effective patterns contribute competing attractors.

4.3 The Calibration Gap

The signal chain from multiplicity weights to decoded sequence phenotype passes through several transformations:

$$\rho \xrightarrow{\text{exact}} f_{\text{eff}} \xrightarrow{\text{exact}} \bar{a}_{\text{des}} \xrightarrow{\text{lossy}} f_{\text{soft}} \xrightarrow{\text{nonlinear}} f_{\text{obs}}, \tag{5}$$

where $\bar{a}_{\text{des}}$ is the mean attention weight on designated patterns during sampling, f_{soft} is the continuous-valued residue probability at the marker position, and f_{obs} is the fraction of generated sequences displaying the designated phenotype after argmax decoding. We define the calibration gap $\Delta = f_{\text{eff}} - f_{\text{obs}}$ and decompose it into three independently measurable components:

$$\Delta = \underbrace{(f_{\text{eff}} - \bar{a}_{\text{des}})}_{\Delta_{\text{attn}} \approx 0} + \underbrace{(\bar{a}_{\text{des}} - f_{\text{soft}})}_{\Delta_{\text{PCA}}} + \underbrace{(f_{\text{soft}} - f_{\text{obs}})}_{\Delta_{\text{argmax}}}. \tag{6}$$

To predict the PCA gap across families, we define the Fisher separation index $S = (\bar{c}_{\text{within}} - \bar{c}_{\text{between}})/[\frac{1}{2}(\sigma_{\text{within}} + \sigma_{\text{between}})]$, where $\bar{c}$ and σ denote the mean and standard deviation of pairwise cosine similarities within and between functional groups in PCA space. For each experiment, we followed the stochastic attention pipeline of [1] without modification to the core sampler. Seed alignments were downloaded from InterPro/Pfam in Stockholm format. Alignments were cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps in the remaining columns. The cleaned alignment was one-hot encoded (20 amino acid channels per position, gaps mapped to zero vectors) yielding a $20L \times K$ matrix, reduced by PCA (retaining 95% of variance) to a $d \times K$ matrix, and column-normalized to unit norm. ULA chains were run with step size $\alpha = 0.01$ for $T = 5{,}000$ steps, initialized near random stored patterns with small Gaussian perturbation. Samples were collected after a burn-in of 2,000 steps with thinning every 100 steps. For multiplicity-weighted runs, $\beta^*(\mathbf{r})$ was found by sweeping 50–60 logarithmically spaced β values in $[0.1,\ 500]$ and locating the inflection point of the weighted attention entropy.

Protein families and functional splits. We tested three Pfam families: Kunitz domains (PF00014, $K = 99$, $L = 53$), SH3 domains (PF00018, $K = 55$, $L = 48$), and WW domains (PF00397, $K = 420$, $L = 31$). For each family, we defined a binary functional split based on a single marker position. For Kunitz, the P1 position—the primary contact residue with serine proteases, known from structural and biochemical studies to determine inhibitory specificity—was identified as the column with the highest Lys frequency; sequences with K or R at P1 were designated as the functional subset ($K_{\text{des}} = 32$) based on their known association with trypsin inhibition. For SH3, the conserved Trp in the hydrophobic binding groove was identified as the column with intermediate Trp frequency ($0.15 < f_W < 0.85$); sequences with W formed the designated subset ($K_{\text{des}} = 33$). For WW, the specificity-determining position was the highest-entropy column in the middle third of the alignment; sequences with the most common residue formed the designated subset ($K_{\text{des}} = 69$). In each case, the designation reflects external biological knowledge, not a property discovered by the method.

Multiplicity-weighted sampling. For each ρ value, the multiplicity vector was set to $r_k = \rho$ for designated patterns and $r_k = 1$ for background patterns. Weighted samples were generated using the modified Langevin update (Eq. 3). Twenty to thirty independent chains were run per condition, yielding 620–930 sequences after burn-in and thinning.

Evaluation metrics. Phenotype fidelity was measured as the fraction of generated sequences displaying the designated subset's marker residue at the marker position. Attention diagnostics recorded the mean softmax weight on designated patterns during sampling. The soft P1 score was computed by reconstructing the PCA-space sample to one-hot space and applying a temperature-scaled softmax ($T = 0.2$) to the amino acid channels at the marker position. Sequence diversity was computed as one minus the mean pairwise sequence identity among generated sequences (estimated from 300–500 random pairs). Amino acid composition fidelity was measured by KL divergence between generated and reference amino acid frequencies.

Fisher separation index. For each functional split, pairwise cosine similarities were computed between all unit-norm PCA vectors within Group A, within Group B, and between groups. The pooled within-group mean $\bar{c}_{\text{within}}$ and standard deviation σ_{within} were computed by weighting by group size; the separation index S was then computed as defined in Section 4.3.

Software. All experiments were implemented in Julia 1.12 using NNlib.jl (softmax), MultivariateStats.jl (PCA), and Distributions.jl. Source code and experiment scripts are available at https://github.com/varnerlab/SA-Binding-Generation-Study.

Acknowledgments

Author contributions. J.D.V. conceived the study, developed the theory, wrote the code, performed the experiments, and wrote the paper.

Competing interests. The author declares no competing interests.

Data availability. All Pfam seed alignments are publicly available from InterPro. Source code and experiment scripts are available at https://github.com/varnerlab/SA-Binding-Generation-Study.

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SI Appendix

S1 Supplementary Tables

S2 Supplementary Figures